



ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

College of Engineering

Department of Electrical and computer Engineering

Project proposal on
An SLM-based Socratic Assistant for Student
Material Mastery

Name	ID
1. Anwar Abdela	ETS0207/14
2. Aryam Ezra	ETS0220/14
3. Bamlak Geteye	ETS0247/14
4. Bereket Tadiwos	ETS0214/13

Advisor: Mr. Sileshi A.

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DECLARATION

We, **Anwar Abdela, Aryam Ezra, Bamlak Geteye and Bereket Tadiwos**, fifth year students of Electrical and Computer Engineering at Addis Ababa Science and Technology University (AASTU), hereby declare that this Final Year Project report entitled: “**SLM-based Socratic Assistant for Student Material Mastery**”, is our original work and is based on the knowledge, experience, and findings we have gained throughout our study period. All the contents included in this proposal are genuine, and any external sources used have been properly acknowledged and referenced in accordance with academic standards.

We affirm that this report has not been submitted to any other institution for the award of any academic qualification. We also acknowledge the continuous support, guidance, and mentorship provided by our advisor throughout the development of this project.

Therefore, we respectfully submit this proposal in partial fulfillment of the requirements for the Bachelor of Science Degree in Electrical and Computer Engineering.

Advisor's Approval

This certifies that the above declaration is correct and that this Final Year Project has been carried out under my supervision and is hereby approved for submission.

Name: Mr. Sileshi A.

Signature: _____

Date: _____

Final Year Project Coordinator Approval

This report has been reviewed and approved for submission as part of the requirements for the Bachelor of Science Degree.

Name: _____

Signature: _____

Date: _____

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence

API - A formal Application Programming Interface

BAAI - Beijing Academy of Artificial Intelligence

BERT - Bidirectional Encoder Representations from Transformers

BM25 - Best Matching 25

CILP - Constructivist Inquiry-Based Learning Prompting

CPU - Central Processing Unit

CVs - Curricula Vitae

DB - Database

FAISS - Facebook AI Similarity Search

FERPA - Family Educational Rights and Privacy Act

GDPR - General Data Protection Regulation

GGUF - GPT-Generated Unified Format

HTTP - Hypertext Transfer Protocol

ICCCA - International Conference on Computing, Communication and Automation

ITS - Intelligent Tutoring Systems

JSON - JavaScript Object Notation

KLI - Knowledge-Learning-Instruction

KM - Knowledge Management

LaTeX - Lamport TeX

LLMs - Large Language Models

MRR - Mean Reciprocal Rank

MTEB - Massive Text Embedding Benchmark

NLP - Natural Language Processing

OCR - Optical Character Recognition

PDF - Portable Document Format

PoC - Proof of Concept

RAG - Retrieval-Augmented Generation

SLM - Small Language Model

STEM - Science, Technology, Engineering, and Mathematics

UI - User Interface

UX - User Experience

CHAPTER ONE

INTRODUCTION

The integration of abstract theoretical knowledge with practical applications continues to present significant challenges in STEM education, particularly in physics and mathematics. Research work demonstrates that students frequently struggle to connect mathematical formalism with physical intuition, resulting in limited problem-solving transferability and conceptual understanding [1]. The usual traditional educational approaches and existing digital learning platforms fail to provide adequate personalized scaffolding, offering standardized content that cannot adapt to individual learning trajectories or domain-specific contexts.

This project aims to bridge these limitations through an SLM-based Socratic Assistant implementing Retrieval-Augmented Generation (RAG) architecture, building upon recent advancements in educational AI systems [2]. The system constructs a dynamic knowledge base from individual students' academic materials including lecture notes, textbooks, and problem-solving histories, enabling context-aware tutoring that emulates the adaptive support mechanisms of human experts [3]. By embedding established pedagogical frameworks that prioritize conceptual understanding and real-world applications, the assistant delivers personalized guidance across fundamental undergraduate topics, transforming passive content consumption into an active learning partnership that cultivates both technical proficiency and practical intuition.

1.1 Background of the Study

Here are the key technical concepts involved in the project, explained to create the contextual background necessary to understand the project.

Static vs. Dynamic Knowledge Repositories

Static repositories are centralized storages for archival and retrieval purposes. They require users to know exactly what they are searching for, lack contextual understanding of queries, cannot adapt to individual learning needs, and provide no guidance on how to apply retrieved information. Dynamic KM systems address these shortcomings through intelligent query interpretation, personalized content recommendation, adaptive learning pathways, and integration of multiple knowledge sources.

AI-Enhanced Knowledge Management Architecture

Modern AI-powered KM systems incorporate several key components: semantic understanding engines that interpret user intent, knowledge graphs that map relationships between concepts, recommendation algorithms that suggest relevant materials, and feedback loops that continuously improve system performance based on user interactions.

These KM systems can also be implemented for educational purposes as well. The usual method of education has a slow pace of development and is characterized with deprived smart KM system. Studies show AI based KM implementation in the higher education sector has a promising result.

Adaptive Learning Technologies

Current adaptive learning systems employ various approaches: rule-based systems that follow predetermined branching logic, Bayesian knowledge tracing that models student understanding probabilistically, item response theory that adjusts difficulty based on performance, and deep learning models that identify patterns in student behavior. Each approach has strengths and limitations in terms of accuracy, interpretability, and computational requirements.

RAG Architecture Fundamentals

Retrieval-Augmented Generation represents a hybrid approach combining information retrieval with text generation. Unlike pure language models that rely solely on parameters learned during training, RAG systems retrieve relevant information from external knowledge bases before generating responses. This architecture consists of a retriever component that finds relevant documents, an encoder that processes retrieved information, and a generator that produces coherent responses grounded in retrieved content. Recent systematic surveys summarize RAG's technical components, deployment patterns, and practical deployment challenges, reinforcing RAG as the recommended architecture for knowledge-intensive tutoring systems. A comprehensive review of RAG explains core pipeline stages—chunking, dense embedding, (re)ranking, and generator fusion strategies (early/late/weighted)—and formalizes RAG as marginalizing generator outputs across retrieved documents, which explains how grounding reduces hallucinations and allows knowledge updates without retraining [6].

Vector Database Technologies

Vector databases provide efficient storage and retrieval of embeddings. Technologies like FAISS, Pinecone, Weaviate, and Milvus offer different trade-offs between speed, accuracy, and scalability. Key features include approximate nearest neighbor search for fast retrieval, filtering capabilities to constrain searches, and hybrid search combining semantic and keyword approaches.

1.2 Problem Statement

Have you ever finished a course and realized you didn't truly understand the main ideas? Have you ever looked back and asked yourself: *Did I actually learn anything? Why am I learning this? Where is this knowledge useful, and why should I study something I won't use in my day-to-day life?*

Many students face these questions because there is a significant gap between theoretical knowledge, real understanding, and the ability to apply that knowledge in real-world situations.

Research demonstrates a documented gap between academic learning and practical application. Globally, surveys reveal that more than **60%** of employers believe graduates lack real-world experience, and **51%** believe graduates do not have the right skill sets required for jobs. Moreover, **85%** of graduates feel their degrees didn't prepare them for workplace demands, and just **24%** say they have all the skills they need for their current role [10].

In Ethiopia, studies report that **80%** of students reported that their academic programs lacked sufficient practical experience such as internships, projects, or fieldwork. **75%** of employers expressed concern that graduates lacked the practical problem-solving skills necessary for effective performance in the workplace, and **68%** of students felt they were not adequately prepared to face professional challenges upon graduation — contributing to ongoing dependence on foreign experts for technical roles that local graduates should be filling [11].

Our SLM-based Assistant is designed to bridge this gap. It serves as a specialized tool trained on STEM-related subjects that can accurately analyze the information students provide, identify misunderstandings, and guide them toward clear comprehension and practical application of key concepts.

By leveraging a RAG-based model, the assistant minimizes hallucinations and ensures reliable, context-aware feedback. Ultimately, our goal is to help students move from confusion to clarity—enabling them to understand concepts deeply, apply them effectively in real-world scenarios, and build meaningful experience and skills they can confidently showcase on their CVs.

1.3 Objective

1.3.1 General Objective

To design and develop an SLM-based Socratic Assistant to manage personal knowledge that bridge theoretical knowledge to the practical world

1.3.2 Specific Objectives

- To investigate and identify the core problem of the traditional knowledge management and educational system
- To design an AI integrated solution targeting the identified problems
- To develop an initial proof of concept for the solution
- To deploy a fully functional system at a prototype level
- To perform various tests on the proposed system

1.4 Significance of the Study

This project holds significant value for the academic community and the field of educational technology by directly addressing the **conceptual application deficit** in STEM education.

1.4.1 Bridging the Theoretical-Practical Dichotomy

Current educational research indicates that a majority of students struggle to connect mathematical formulas and equations with physical intuition, often failing to apply learned formulas to novel contexts shortly after instruction [1]. This study aims to resolve this by creating a tool that transforms passive content consumption into an active learning partnership, facilitating the transition from rote memorization to deep conceptual transfer.

1.4.2 Advancing AI-Enhanced Knowledge Management

While organizations have traditionally relied on static archival repositories for documentation, the integration of AI into these systems is necessary to meet modern standards of efficiency and context-awareness [4]. This project demonstrates a practical application of dynamic Knowledge

Management (KM) in education, moving away from passive storage toward systems that can interpret user intent and synthesize information from personal notes.

1.4.3 Mitigating Generative Inaccuracies

Standard generative models often suffer from "hallucinations" or lack of grounding, which is critical in engineering fields where precision is paramount. By utilizing Retrieval-Augmented Generation (RAG), this project implements a solution that significantly outperforms language-only models in knowledge-intensive tasks, ensuring that responses are grounded in factual, retrieved data [2].

1.4.4 Promoting Deep Learning through Dialogic Scaffolding

The study integrates established pedagogical frameworks into AI interaction. By scaffolding the system to use inquiry-based learning rather than direct answering, the project leverages constructive cognitive effort to foster critical thinking. Research suggests that this type of guided interaction with AI yields significantly better learning outcomes and conceptual understanding than unguided usage [7].

1.5 Scope

The scope defines the technical boundaries, data sources, and functional capabilities of the Personal Knowledge SLM-based Assistant based on the design methodology.

1.5.1 Functional Scope

The system is designed as a web-based application employing a Retrieval-Augmented Generation (RAG) architecture. Its core functions are limited to:

- Ingestion: Converting multi-format files (PDF, LaTeX, Text) into vector embeddings.
- Retrieval: Using semantic search (Cosine Similarity) to locate relevant information chunks within the database.
- Generation: Producing context-aware responses using a Small Language Model (SLM) configured to follow Socratic tutoring principles.

1.5.2 Data Scope

To ensure the system functions as a reliable academic tutor, the knowledge base is strictly limited to a "Closed-Domain" environment. The data ingestion pipeline is designed to process the following specific academic formats:

- **Primary Unstructured Data:** Digital lecture slides (PDF) and reference textbooks officially utilized by the AASTU Department of Electrical and Computer Engineering.
- **Structured STEM Data:** LaTeX (.tex) source files. These are prioritized because they preserve the semantic structure of complex mathematical equations and symbols better than standard optical character recognition (OCR).
- **Semi-Structured Data:** Historic problem sets, solution keys, and laboratory manuals.

Justification for Data Selection

The restriction of the Retrieval-Augmented Generation (RAG) system to these verified local files serves two critical engineering objectives:

- **Curriculum Alignment:** The system's responses are forced to align with the specific methodologies, notations, and definitions taught within the university curriculum, avoiding generic or conflicting information often found in open-domain Large Language Models.
- **Hallucination Mitigation:** By grounding the generation process exclusively in verified textbooks, the probability of the model fabricating information is significantly minimized.

1.5.3 Methodological Scope

The development follows a Modular Systems Architecture with two parallel streams: the AI Backend (RAG logic, Vector DB) and the Full-Stack Application (UI/UX), as outlined in the project's design methodology. Validation is scoped to technical performance (latency, retrieval accuracy) and qualitative student feedback.

1.6 Limitations

While the proposed solution addresses key gaps in educational technology, it is subject to specific limitations inherent to the current state of AI and the chosen architecture.

1.6.1 Dependence on Data Quality

The system's output quality is strictly bound by the input data. If a student uploads poorly structured notes or incorrect material, the RAG system may retrieve and synthesize inaccurate information. This challenge regarding data quality and availability is a known hurdle in implementing AI in knowledge management systems [5].

1.6.2 Privacy and Security Risks

Since the system indexes personal student documents, which may contain sensitive data, privacy remains a significant concern. While the project aims for local or secure processing, the deployment of RAG systems on user-owned data requires robust privacy controls and consent workflows that are complex to fully implement in a prototype phase [6].

1.7 Project flowchart

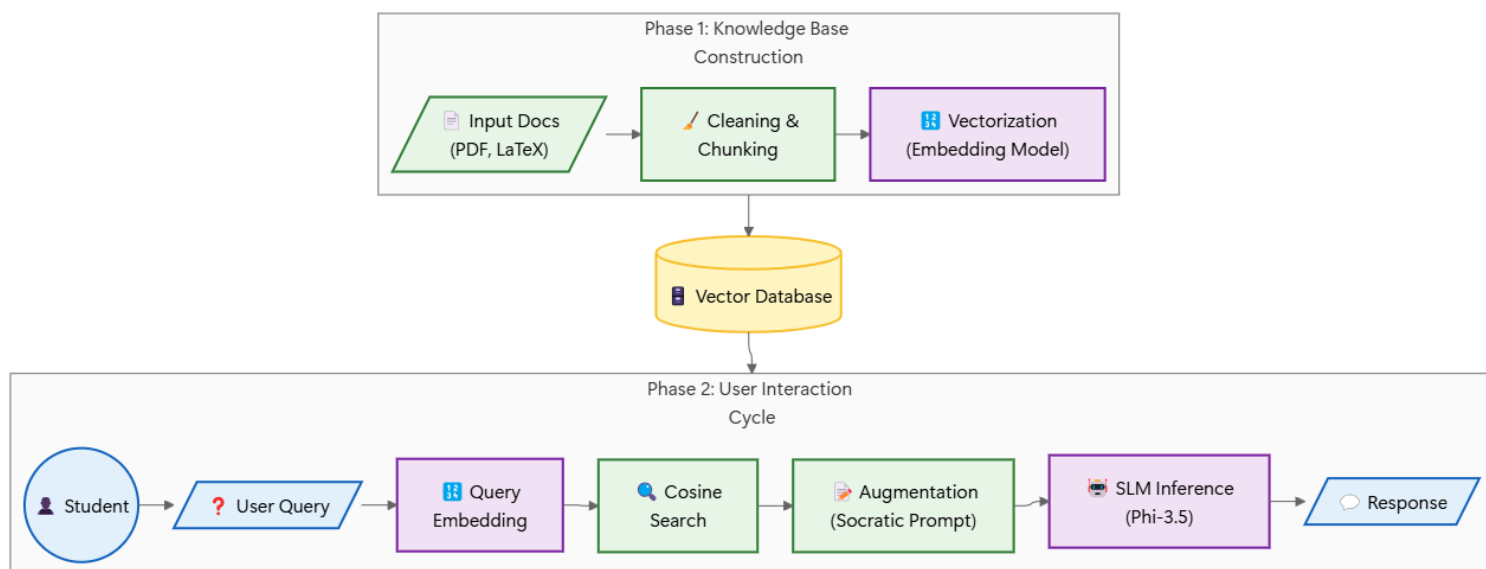


Figure 1: Flowchart of the proposed system

CHAPTER TWO

LITERATURE REVIEW

2.1 Knowledge Management Systems in Education

Companies rely on documentation to expand and utilize their knowledge base. They can achieve this in two methods, either with static repositories or dynamic Knowledge Management (KM) systems. To meet today's standard, the second option is most preferred. There are simple dynamic KMs that have been around for a while but empowering them with AI boosts the efficiency. This is a suitable implementation of AI in knowledge management [4].

Evolution of Knowledge Management Systems

The trajectory of KM systems has progressed through three distinct generations. First-generation systems (1990s-2000s) focused on document management and basic search functionality, essentially serving as digital filing cabinets. Second-generation systems (2000s-2010s) introduced taxonomies, metadata tagging, and collaborative features, enabling better organization and knowledge sharing. Third-generation systems (2010s-present) leverage artificial intelligence, machine learning, and natural language processing to provide intelligent, context-aware knowledge delivery.

Challenges of implementing AI in existing system

The implementation of AI in the knowledge management system face many challenges. One of the main challenges in implementing AI into such knowledge management systems is the question of privacy. The data used to train these models may involve personal and organizational data. There is also the concern of biasing in the training process. If the training is biased, it compromises the integrity and reliability of the implemented model to do its task.

As a new system, integration of AI into an organizational system must align with the organizational value and principles. Defying these values results in resistance to embrace the technological significance of the new system. Historically, new technologies of their era also

have technological challenges regarding the available resources, security, robustness and complexity.

All these concerns regarding AI implementation into an existing system can be classified into three major groups. Technological challenges: includes factors like data quality and availability, scalability, transparency, reliability and robustness, security and privacy, and algorithm complexity, organizational challenges: involves factors like resistance to change, lack of expertise, interdepartmental collaboration, and budget constraints and ethical challenges: encompassing the factors bias, accountability, privacy, transparency, and lack of job security [5].

2.2 The Learning Crisis in STEM Education

Quantifying the Problem

The integration of abstract theoretical knowledge with practical applications continues to present significant challenges in STEM education, particularly in physics and mathematics. Current research demonstrates that students frequently struggle to connect mathematical formalism with physical intuition, resulting in limited problem-solving transferability and conceptual understanding.

Wieman's comprehensive study revealed alarming statistics: only 30% of students retain core concepts one semester after instruction, 65% of students cannot apply learned formulas to novel problems, 45% report feeling "lost" during lectures despite attending regularly, and 70% rely on memorization rather than understanding. These findings suggest fundamental flaws in the usual educational approaches.

Cognitive Load Theory and STEM Learning

Cognitive load theory provides insight into why students struggle with complex STEM topics. Working memory has limited capacity, and when instructional materials impose excessive cognitive load, learning suffers. STEM subjects often present high intrinsic complexity (inherent difficulty of the material), extraneous load (poor instructional design), and insufficient germane load (cognitive resources devoted to schema construction and automation).

The Transfer Problem

One of the most significant challenges in STEM education is the transfer of learning—the ability to apply knowledge learned in one context to new situations. Research indicates that students often develop "brittle knowledge" that works only for problems identical to those practiced. They struggle to recognize when a concept applies to a novel scenario, cannot adapt solution strategies to new problem types, and fail to connect related concepts across different courses.

Misconceptions and Conceptual Change

Students enter STEM courses with pre-existing mental models, often based on everyday experience, that conflict with scientific understanding. These misconceptions are remarkably resistant to change. Traditional instruction that simply presents correct information rarely succeeds in overcoming deeply held incorrect beliefs. Effective learning requires explicitly confronting misconceptions, providing cognitive conflict that challenges existing beliefs, and offering alternative frameworks that better explain observations.

2.3. Personalized Learning and Adaptive Systems

The usual educational approaches and existing digital learning platforms fail to provide adequate personalized scaffolding, offering standardized content that cannot adapt to individual learning trajectories or domain-specific contexts.

The Case for Personalization

Educational research consistently demonstrates that personalized instruction significantly outperforms one-size-fits-all approaches. Students learn at different rates, require different amounts of practice, benefit from different explanation styles, and have varying levels of prior knowledge. Personalized learning systems can adjust content difficulty dynamically, provide additional examples when needed, skip material already mastered, and adapt explanation style to student preferences.

Intelligent Tutoring Systems (ITS)

Intelligent Tutoring Systems represent a mature approach to personalized education. Successful ITS implementations share common features: detailed student models tracking knowledge and skills, expert models encoding domain knowledge, pedagogical models determining instructional strategies, and user interfaces facilitating natural interaction. Studies show well-designed ITS can approach the effectiveness of human one-on-one tutoring.

The review also shows that targeted (guided) use of generative AI yields better learning outcomes than unguided frequent usage, indicating that personalization must be coupled with scaffolding on how to interact with the system. This suggests the personal assistant should include onboarding / prompt templates and scaffolded workflows (e.g., “explain concept”, “Socratic hint”, “worked numerical example”) rather than leave all prompting to novice users [7].

Limitations of Current Platforms

Despite advances, most educational technology platforms exhibit significant shortcomings. They offer limited content coverage restricted to specific topics, lack integration with students' personal materials, provide superficial feedback without deep explanation, cannot handle open-ended questions effectively, and fail to develop metacognitive skills and learning strategies.

2.4 Retrieval-Augmented

Generation: A Paradigm Shift

This project addresses these limitations through a SLM-based Socratic Assistant implementing Retrieval-Augmented Generation (RAG) architecture, building upon recent advancements in educational AI systems.

Advantages Over Pure Language Models

RAG systems address critical limitations of standard language models. They reduce hallucinations by grounding responses in factual sources, enable easy knowledge updates without retraining, provide transparency through source attribution, and scale to large knowledge bases

efficiently. For educational applications, these advantages are particularly valuable as accuracy and verifiability are paramount.

RAG in Knowledge-Intensive Tasks

Lewis et al. demonstrated RAG's superiority in knowledge-intensive NLP tasks, showing improvements of 10-20% over comparable language-only models on question answering benchmarks. The system performs particularly well on tasks requiring factual accuracy, integration of multiple information sources, and handling of rare or specialized knowledge.

Educational RAG Implementations

Applying RAG to education requires specific adaptations. Educational RAG systems must handle multi-modal content (text, equations, diagrams), maintain pedagogical appropriateness in responses, balance accuracy with appropriate difficulty level, and support progressive disclosure of information. Recent implementations have shown promising results in mathematics tutoring, science education, and programming instruction.

Technical Challenges in Educational RAG

Several technical challenges arise when implementing RAG for education: chunking strategies must preserve conceptual coherence, retrieval must consider pedagogical relevance not just semantic similarity, generation must follow Socratic principles rather than direct answer provision, and the system must handle mathematical notation and equations effectively.

2.5 The Socratic Method and AI-Powered Tutoring

The system constructs a dynamic knowledge base from individual students' academic materials including lecture notes, textbooks, and problem-solving histories, enabling context-aware tutoring that emulates the adaptive support mechanisms of human experts.

Historical Foundation of Socratic Teaching

The Socratic method, developed by the ancient Greek philosopher Socrates, emphasizes learning through questioning rather than direct instruction. This approach guides students to discover knowledge independently, develops critical thinking skills, exposes gaps in understanding, and

promotes deeper, more durable learning. Modern educational psychology has validated the effectiveness of this approach, particularly for developing higher-order thinking skills.

Cognitive Science Foundations

The Knowledge-Learning-Instruction (KLI) framework proposed by Koedinger et al. provides theoretical grounding for effective tutoring. This framework distinguishes between different types of knowledge: declarative knowledge (facts and concepts), procedural knowledge (how to perform tasks), and conditional knowledge (when to apply procedures). Effective instruction must address all three types and facilitate their integration.

Desirable Difficulties and Productive Struggle

Contemporary learning science emphasizes that some difficulty during learning is beneficial. "Desirable difficulties" that require effortful retrieval, generation of answers, and connection-making lead to stronger learning than passive review. Socratic tutoring naturally incorporates these principles by requiring students to actively construct understanding rather than passively receive information.

Implementation in AI Systems

Implementing Socratic tutoring in AI systems requires sophisticated prompt engineering and dialogue management. The system must ask appropriate questions that guide without revealing answers, recognize when students are genuinely stuck versus productively struggling, provide hints at appropriate levels of specificity, and adjust questioning strategy based on student responses. Multi-turn dialogue capabilities and context maintenance are essential.

Prompt Engineering for Socratic Dialogue

Effective Socratic AI tutors employ multi-layered prompts that establish the tutoring role and pedagogical approach, incorporate retrieved context from student materials, structure the dialogue to guide discovery, and include safeguards against providing direct answers. Prompt design must balance guidance with autonomy, ensuring students remain engaged without becoming frustrated.

A controlled classroom study validates a constructivist prompt-engineering approach called the Constructivist Inquiry-Based Learning Prompting (CILP) framework, where students were trained to craft, refine, and critique prompts when using ChatGPT as a supplemental tutor. The intervention (n=120, quasi-experimental) demonstrated statistically significant gains in conceptual understanding and reductions in qualitative misconceptions (e.g., entropy, internal energy) when ChatGPT was used in a guided, inquiry-based manner — though quantitative/numerical problem-solving gains were limited. This supports embedding explicit prompt-crafting and metacognitive reflection into AI tutoring workflows [7].

2.6 Vector Databases and Semantic Search Technology

The usual information retrieval based on keyword matching suffers from vocabulary mismatch problems. Students may phrase questions differently than how information appears in source materials, synonyms and related terms are not recognized, and conceptual relationships between topics are ignored. Semantic search addresses these limitations by representing text based on meaning rather than exact words.

Vector Embeddings and Semantic Space

Modern semantic search relies on vector embeddings—dense numerical representations of text in high-dimensional space. These embeddings are learned by neural networks trained on large text corpora, capturing semantic relationships such that similar concepts have similar vector representations. In this semantic space, mathematical operations correspond to meaningful relationships: similar vectors represent similar concepts, vector arithmetic can capture analogies (e.g., "king" - "man" + "woman" $\approx$ "queen"), and distance metrics quantify semantic similarity.

Embedding Models for Educational Content

Choosing appropriate embedding models is crucial for educational applications. General-purpose models like Sentence-BERT provide strong baseline performance, domain-specific models fine-tuned on scientific text better capture technical terminology, and multi-modal models can handle equations and diagrams alongside text. Recent research explores embedding models specifically optimized for educational content, incorporating pedagogical relationships between concepts. The review highlights practical chunking strategies specific to educational content

(semantic chunking that respects topic boundaries, controlled overlap to retain continuity, and metadata enrichment) and recommends hybrid retrieval (dense + BM25) plus re-ranking for increased precision in high-stakes domains. These findings directly inform index construction for student-uploaded materials. [6]

Chunking Strategies for Educational Content

Effective RAG systems require careful text chunking—dividing documents into segments for embedding and retrieval. Educational content presents unique challenges: preserving complete explanations and derivations, maintaining context for mathematical expressions, balancing granularity (small chunks for precision vs. large chunks for context), and handling hierarchical structure (sections, subsections, examples).

Research suggests optimal strategies include semantic chunking that respects topic boundaries, overlapping chunks (10-15%) to prevent context loss, metadata enrichment with section titles and topics, and adaptive chunk sizing based on content density.

2.7 Multi-Document Integration and Context Synthesis

The Challenge of Fragmented Knowledge

Students' learning materials are inherently fragmented across multiple sources: lecture notes from different sessions, textbook chapters, supplementary readings, problem sets, and personal notes. Traditional search returns individual fragments without synthesizing information across sources. Effective learning requires integrating these fragments into coherent understanding.

Cross-Document Reasoning

Advanced RAG systems must perform cross-document reasoning: identifying complementary information across sources, resolving contradictions or different perspectives, synthesizing comprehensive answers from multiple fragments, and tracking provenance to attribute information to sources. This capability is particularly valuable for complex topics covered differently in various materials.

Context Window Management

Language models have limited context windows—the amount of text they can process simultaneously. When retrieved content exceeds this limit, sophisticated strategies are needed: ranking retrieved chunks by relevance, compressing or summarizing less critical information, iterative refinement where initial responses are enhanced with additional context, and hierarchical processing that first identifies relevant documents then focuses on specific sections.

2.8 Evaluation Metrics for Educational AI Systems

Technical Performance Metrics

Evaluating RAG systems requires multiple technical metrics. Retrieval accuracy measures whether the system finds relevant information: precision (proportion of retrieved chunks that are relevant), recall (proportion of relevant chunks that are retrieved), and Mean Reciprocal Rank (MRR) assessing ranking quality. Generation quality evaluates the produced responses: factual accuracy against source materials, coherence and fluency of explanations, and faithfulness to retrieved content without hallucination.

Pedagogical Effectiveness Metrics

Beyond technical performance, educational systems must be evaluated on learning outcomes: conceptual understanding measured through pre- / post-testing, transfer ability to solve novel problems, retention over time, and metacognitive development including learning strategies and self-assessment accuracy.

The RAG survey further synthesizes recommended evaluation axes—retrieval metrics (precision/recall/MRR), generation faithfulness (fact-check / hallucination rates), latency/compute efficiency, and task-specific pedagogical outcomes — recommending mixed technical + pedagogical evaluation for educational RAG systems. These metrics should be used to evaluate both your retriever (how often it finds the right passage) and generator (how accurately it explains or scaffolds learning) [6].

User Experience Metrics

Student satisfaction and engagement are critical for adoption: perceived usefulness and ease of use, time spent with the system, types of questions asked, and qualitative feedback on explanation quality.

2.9 Ethical Considerations and Limitations

Academic Integrity Concerns

AI tutoring systems must carefully balance support with academic integrity. Systems should guide learning without completing assignments, encourage understanding over answer-seeking, and include safeguards against misuse. Transparent communication about appropriate use cases is essential.

Bias and Fairness

AI systems can perpetuate biases present in training data or source materials. Educational AI must be evaluated for equitable performance across different student populations, representation of diverse perspectives, and accessibility for students with disabilities.

Privacy and Data Security

Student data is highly sensitive. Systems must implement robust security measures, obtain informed consent, provide transparency about data usage, and comply with regulations like FERPA and GDPR. The review explicitly flags privacy and security as open challenges for RAG when indexing user-owned or proprietary data, and points to emerging directions (privacy-preserving RAG, federated vector DBs, access control layers) that must be considered when building systems that index student notes. For a personal assistant that indexes student documents, implementing privacy controls, consent workflows, and possible on-device or federated indexing will be necessary to mitigate risk [6].

Limitations of AI Tutoring

AI systems cannot fully replace human instructors. They lack true understanding and may miss subtle student difficulties, cannot provide emotional support and motivation, and may struggle

with highly creative or open-ended problems. AI tutoring is best viewed as a complement to, not replacement for, human instruction.

2.10 Related Works, Gap Analysis and Project Contribution

Current Landscape Limitations

Despite significant advances in educational technology and AI, substantial gaps remain in available solutions. Existing systems typically focus on single domains with preloaded content, lack integration with students' personal materials, provide limited personalization based on individual learning trajectories, offer superficial feedback without deep Socratic guidance, and fail to synthesize information across multiple sources.

While CILP and other generative-AI tutoring interventions demonstrate clear benefits for conceptual understanding, they also expose key limitations: generative models without retrieval grounding can struggle with numerical accuracy and with citing sources. This underscores our choice to use a RAG-based assistant that fuses the dialogic strengths of LLMs (Socratic questioning, conceptual scaffolding) with retrieval grounding (student materials, worked examples and authoritative sources) to reduce hallucinations and improve numeric/derivation accuracy [7].

This Project's Unique Contribution

This Personal SLM-based Socratic Assistant addresses these gaps through several innovations: personal knowledge base construction from student-uploaded materials, RAG architecture ensuring factual accuracy grounded in student's own content, Socratic tutoring approach promoting deep understanding, multi-document synthesis for comprehensive answers, and context-aware responses considering student's learning history.

The system represents a significant advance toward truly personalized, adaptive STEM education that bridges the gap between theoretical knowledge and practical application, transforming passive content consumption into active learning partnerships.

CHAPTER THREE

DESIGN METHODOLOGY

To address the complexity of integrating artificial intelligence with a functional user application, this project adopts a Modular Systems Architecture supported by a Parallel Development Strategy.

This methodology structure divides the overall system into well-defined, independent modules that can be developed, tested, and improved separately before being integrated into a single complete system. The whole process combines multiple components of technologies to work together, serving for singular purpose. These components need a thorough development process. The modularization technique allows individual development without compromising each other. This smooths out the entirety of the project flow. It also enables iterative feedback loops of development, while minimizing overall project delays.

Here follows the system development structure divided into phases.

3.1 Phase I: Requirement Analysis and Specification

The first phase defines the exact objectives, limitations and expected performance of the system. The focus is on translating the project idea into clear technical specifications. Following the standards of Software requirement and specification, here are the functional and non-functional requirements.

Functional Requirements

The system must be able to:

- Accept multi-format learning materials such as PDF files, LaTeX documents, and plain text.
- Reformat the input data or materials to a machine understandable data
- Perform semantic search, meaning that the system understands the meaning of a question rather than just matching keywords.
- Utilize the existing knowledge base to retrieve relevant information

- Generate context-aware, explanatory responses using a Socratic tutoring approach (guiding the learner through reasoning instead of only giving final answers).
- Allow the user to interact through a web-based chat interface.

Non-Functional Requirements

- Low latency: The system should return results with minimal delay to support natural interaction.
- Scalability: The design should support the addition of new documents without reducing performance.
- Security and privacy: User-uploaded documents and queries must be protected from unauthorized access.
- Usability: The system should be simple enough for students with basic computer knowledge to use easily.

In addition to the specifications, to promote the parallel development structure, early API definition is implemented in this phase as well. Detail explanation of the API definition is provided below.

API Contract Definition

A formal Application Programming Interface (API) contract is defined at this stage. This API specifies how the Frontend (user interface) and Backend (AI + database) will communicate, including:

- Request formats (JSON)
- Response formats
- Required variables and data types

3.2 Phase II: Data Pipeline Engineering

This phase focuses on transforming raw, unstructured educational material into a structured and searchable knowledge base. It forms the foundation of the Retrieval-Augmented Generation (RAG) system.

Data Ingestion and Preprocessing

- Documents such as lecture notes, textbooks, and research materials are loaded into the system.
- Special characters, incorrect encodings, and irrelevant symbols are removed.
- Mathematical equations and formulas are preserved for engineering accuracy.
- Text is normalized into a consistent, readable format.

Semantic Chunking

The cleaned text is divided into smaller, meaningful sections called chunks.

- Each chunk represents a complete idea or concept.
- Overlapping regions (approx. 10%) are used to prevent loss of context between chunks.
- The chunk size is optimized to balance detail and processing efficiency.

Vectorization and Indexing

Each text chunk is converted into a vector embedding using a pre-trained embedding model.

These vectors represent the *semantic meaning* of the text in numerical form.

- The vectors are stored in a Vector Database.
- The database is optimized for Cosine Similarity Search, allowing the system to quickly retrieve the most related pieces of information when a query is made.

This transforms static documents into a searchable semantic knowledge base.

3.3 Phase III: Parallel Development Streams

This phase is executed in two parallel streams to increase efficiency and allow simultaneous progress. Both streams operate independently using the predefined API contract.

Stream A: AI Backend & RAG Engine Development

This stream builds the core intelligence of the system.

RAG Logic Implementation

- The user's question is converted into a query vector.
- The Vector Database is searched for the top k most relevant text chunks.

- These chunks are passed as context to the Small Language Model (SLM).

Model Selection and Computational Optimization

After a comparative analysis of available Small Language Models (SLMs), this project selects **Microsoft's Phi-3.5 Mini**.

Unlike general-purpose models trained on internet scrapings, the Phi series is trained specifically on "textbook-quality" synthetic data. This training methodology results in superior performance on reasoning capabilities and STEM-related logic tasks, making it uniquely improved for an educational tutoring assistant.

The model supports a 128k token context window, allowing the system to ingest and analyze entire chapters or lengthy lecture notes in a single inference pass, ensuring comprehensive answer synthesis.

To balance performance with resource efficiency, the model will be deployed using 4-bit Quantization (GGUF format). This optimization allows for edge deployment, enabling the system to perform low-latency inference on standard computing hardware without reliance on external cloud computing resources or extensive training periods.

For the semantic retrieval component, the system utilizes the **bge-small-en-v1.5 model**. This model was selected for its high placement on the MTEB (Massive Text Embedding Benchmark), offering a high-fidelity vector representation of technical engineering terminology while maintaining a compact computational footprint.

Prompt Engineering

Special multi-layered prompts are designed to:

- Force the AI to act as an engineering tutor
- Follow the Socratic method
- Explain reasoning and practical applications
- Avoid unsupported or imaginary information

Backend Server Construction

- A Backend Server is built to:

- Manage user sessions
- Receive requests from the frontend
- Send queries to the Vector DB and SLM
- Return final responses to the user

Unit Testing

Individual components such as:

- Data retrieval
- Similarity matching
- Prompt generation

are tested independently to ensure correctness.

Stream B: Full-Stack Application Development

This stream builds the visual and interactive part of the project.

UI/UX Design

A clean and simple layout is designed with two main sections:

1. Document Dashboard

- Upload study materials
- View and manage stored files

2. Chat Interface

- Ask questions
- View AI responses
- Support code and equations

Frontend Development

- Includes structured formatting for mathematical outputs

UX Validation

Initial testing is done to ensure:

- Navigation is clear
- Buttons and interactions are user-friendly
- Information is displayed logically

3.4 Phase IV: System Integration and Optimization

This phase merges the two parallel development streams into one functional system. We use the API points we create at the early stage of the project to integrate the modularized partitions of the system.

API Connection and Testing

- The frontend is connected to the backend using the predefined API.
- Full data flow is tested from:
 - User input → AI processing → Displayed response

Performance Optimization

- Latency (response time) is analyzed and reduced where possible.
- Unnecessary data processing is removed to improve speed.
- Memory usage and CPU efficiency are monitored and optimized.

This phase ensures the system behaves as a single coherent unit.

3.5 Phase V: Testing, Validation, and Evaluation

This final phase evaluates the system using both **technical** and **educational** metrics.

Technical Validation

The developers of the system use technical measurements to evaluate the system's final output.

The following key metrics are measured:

- Retrieval Accuracy – How often the correct information is found
- Hit Rate – Whether a relevant document appears in the top results
- MRR (Mean Reciprocal Rank) – Ranking quality of retrieved results
- Fidelity Score – Percentage of answers that directly match the source documents

Pedagogical (Educational) Validation

- Final validation is performed by the end users. The system is tested by undergraduate engineering students. Since the target users are the students, the final validation point is their feedback.
- Feedback is collected on:
 - Clarity of explanation
 - Ability to connect theory to real-world applications
 - Overall usefulness in learning complex topics

This ensures the system is not only technically correct, but also educationally effective.

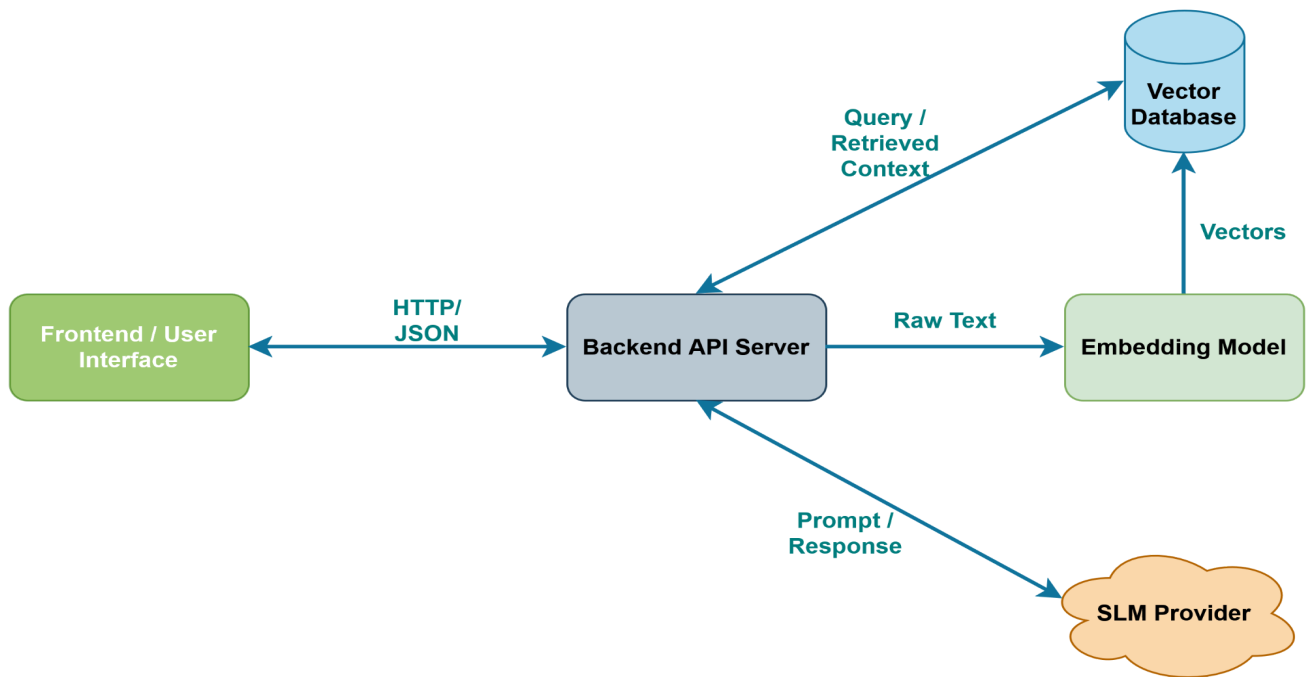


Figure 2: System Architecture and Components Block Diagram of the Proposed System

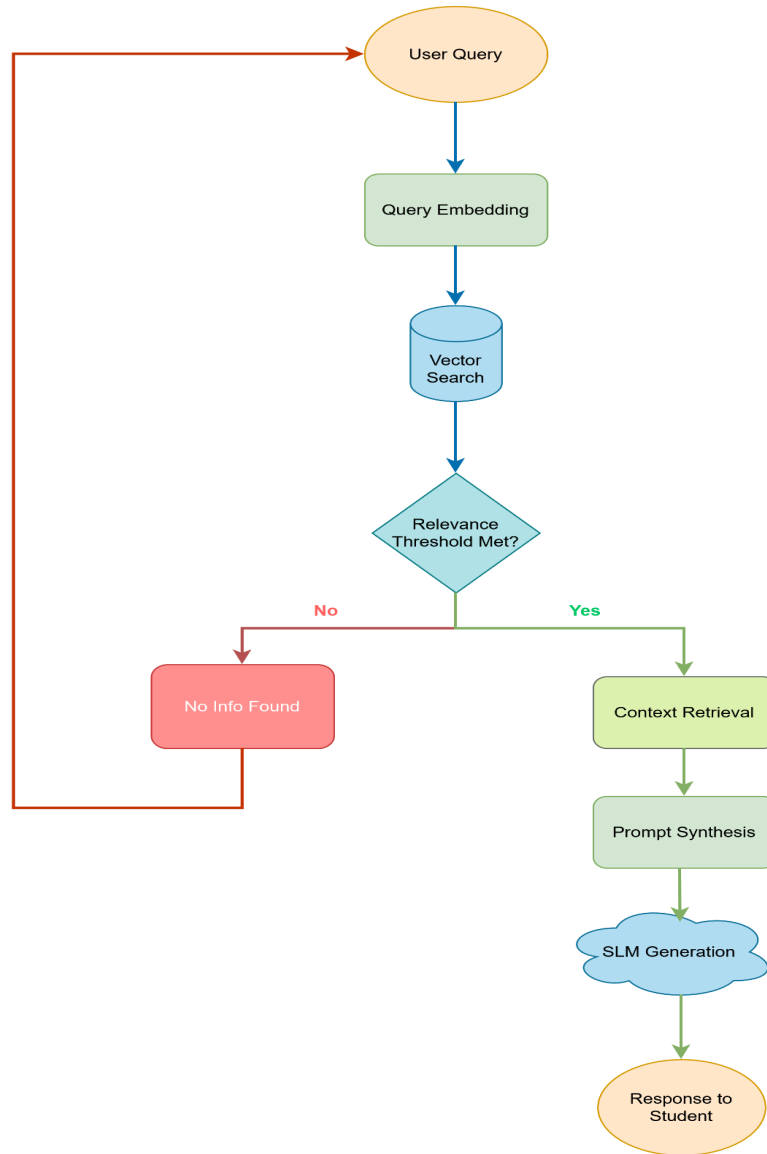


Figure 3: RAG Flow Chart of the Proposed System

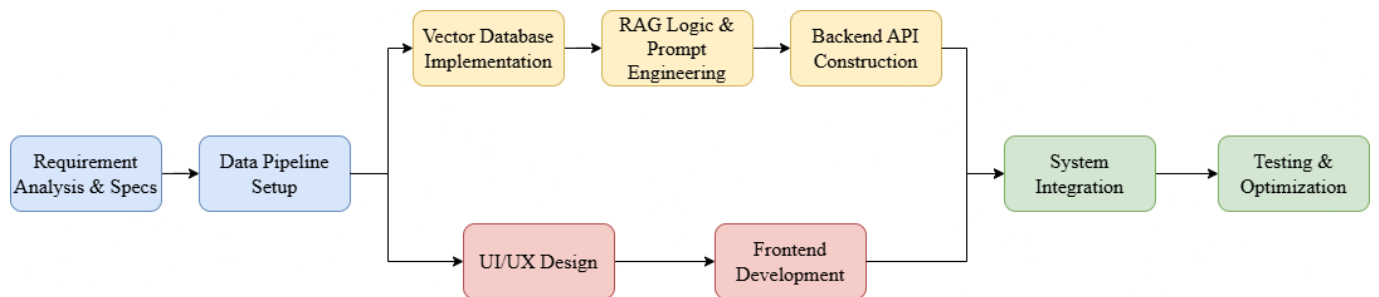


Figure 4: Overall Workflow of the Proposed System

CHAPTER FOUR

WORK PLAN AND TIMELINE

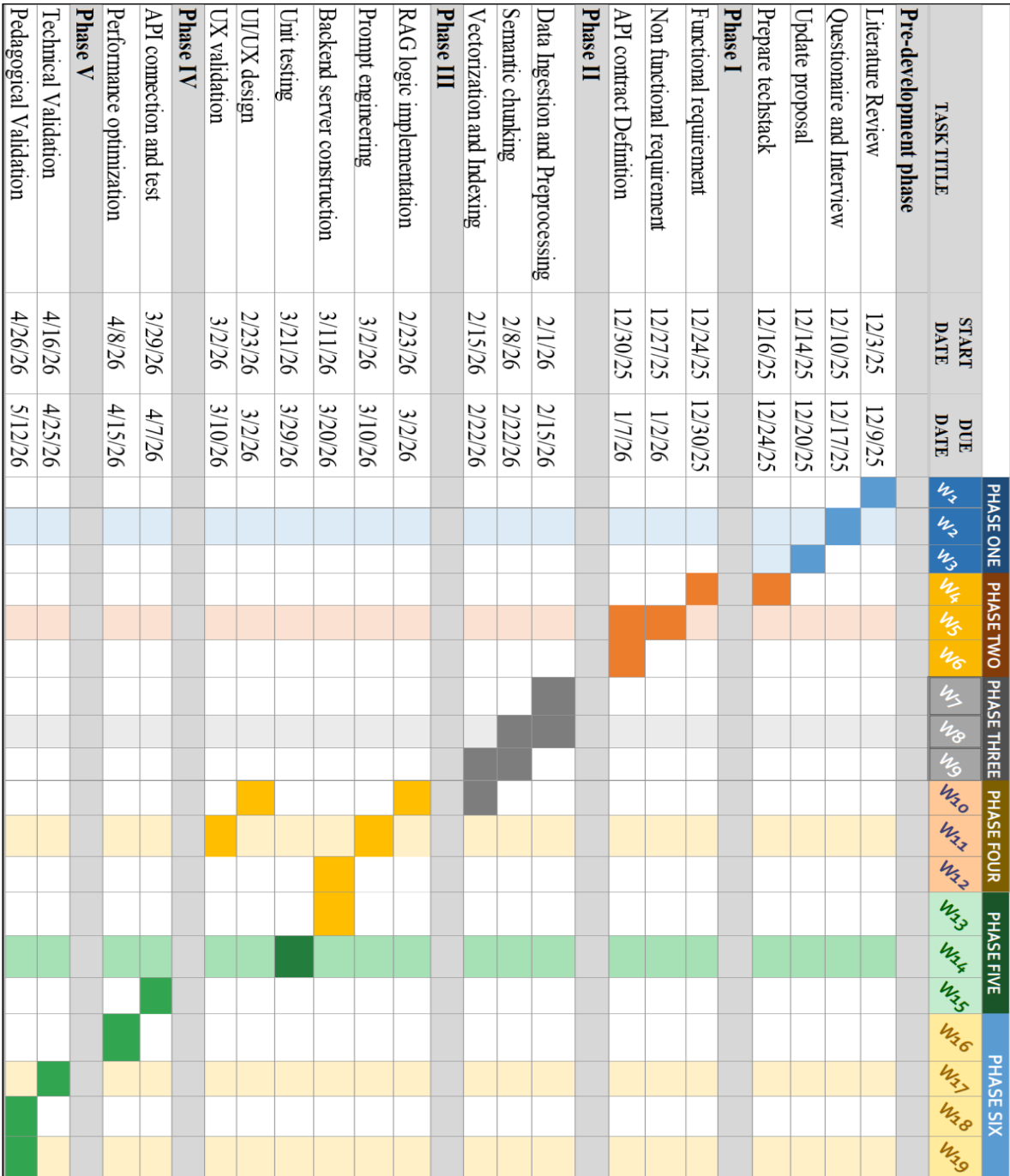


Figure 5: Project timeline

CHAPTER FIVE

EXPECTED RESULTS

By the end, this project aims to achieve the following outcomes:

5.1 Technical Performance

The system is expected to demonstrate superior performance in knowledge-intensive tasks compared to standard language models by reducing fabrication rates. By retrieving relevant document segments before generating answers, the fidelity of the responses to the source material is expected to increase [2].

The search mechanism is expected to successfully overcome semantic discontinuity. Through the use of vector embeddings, the system should correctly identify relevant sections in textbooks or notes even when the student's query uses synonyms or different phrasing than the source text.

5.2 Educational Impact

Students using the system are expected to show improvements in conceptual understanding. By enforcing a guided inquiry mode where the AI asks follow-up questions rather than providing immediate answers, the system aims to replicate the learning gains associated with human one-on-one tutoring [3].

The project expects to validate that targeted, prompt-engineered AI interaction leads to a reduction in common misconceptions in STEM topics, as opposed to the unguided use of generative AI [7].

5.3 Functional Deliverable

The final result will be a working Proof of Concept (PoC) capable of ingesting PDF/LaTeX documents, indexing them into a Vector Database, and allowing a user to interact with this personal knowledge base via a chat interface with low latency and high semantic relevance.

CHAPTER SIX

CONCLUSION

This project was initiated to address the persistent disconnect between abstract theoretical knowledge and practical application in STEM education. Research highlights that traditional educational approaches often fail to facilitate the transfer of learning, leaving students relying on rote memorization rather than deep conceptual understanding [1]. The developed Personal Knowledge SLM-based Assistant serves as a strategic intervention to bridge this gap, offering a specialized tool that transforms passive content consumption into an active, inquiry-based learning partnership.

To achieve this, the system leverages a Retrieval-Augmented Generation (RAG) architecture, moving beyond the limitations of standard generative models. By grounding AI responses in a dynamic knowledge base constructed from the student's own academic materials, the system mitigates the risks of unverified outputs and semantic hallucination commonly associated with Large Language Models [2]. This ensures that the assistance provided is, not only technically accurate but, also contextually aligned with the specific curriculum and lecture notes the student is currently studying.

Beyond technical accuracy, the project integrates established pedagogical frameworks to foster cognitive development. By implementing an inquiry-based tutoring model, the assistant is designed to guide users through reasoning processes rather than providing immediate solutions. This method aligns with cognitive science principles, promoting constructive cognitive effort and helping students identify and correct misconceptions through dialogic scaffolding [7]. This approach addresses the limitations of generic educational platforms that often lack the personalization required to support individual learning trajectories.

Ultimately, this project represents a significant shift from static archival repositories to dynamic, AI-enhanced Knowledge Management systems [4]. By successfully combining semantic search technology with educational theory, the proposed solution offers a viable pathway toward truly personalized education. It demonstrates that AI can serve as a capability amplifier for students,

transforming the solitary act of studying into an interactive, context-aware partnership that effectively prepares them for the complexities of real-world engineering challenges.

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